CONGNING NI

Clinical Informatics | Clinical NLP | Machine Learning for Health | Trustworthy AI

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RESEARCH INTERESTS

- **Longitudinal Clinical Informatics and Risk Prediction**, using EHRs and clinical narratives to construct auditable representations of evolving patient and care states and develop calibrated, interpretable prediction of clinically meaningful transitions.
- **LLM Systems for Clinical and Care Support**, developing retrieval-augmented, role-aware, and context-adaptive systems using clinical NLP and reasoning-focused LLMs.
- **Trustworthy and Responsible AI for Healthcare**, characterizing LLM hallucinations, omissions, bias, and alignment risks in clinical, caregiver-facing, and patient-facing contexts.
- **Reliable Reasoning and Temporal Prediction**, studying confidence–rationale alignment, efficient selective reasoning, evidence-grounded generation, representation privacy, and intervention-conditioned forecasting.
- **Multimodal Clinical Data Science and Biological Aging**, integrating clinical narratives, structured EHRs, imaging-linked cohorts, and biospecimen information for disease and treatment-outcome modeling.
- **Computational Social Dynamics in Digital Health**, modeling online patient and caregiver communication, support-seeking behavior, emotional support, and peer interaction dynamics.

EDUCATION

Ph.D. | Computer Science — Jan 2020 – Aug 2025
Department of Computer Science, Vanderbilt University — Nashville, TN

- Dissertation: *Murmuring of Alzheimer's Care: Analyzing Social Dynamics in Online Alzheimer's Disease and Related Dementias Communities*
- Advisor: Zhijun Yin, Ph.D.
- Committee: Zhijun Yin, Ph.D. (Chair); Bradley Malin, Ph.D.; Lijun Song, Ph.D.; Patricia Commiskey, DrPH; Qingxia Chen, Ph.D.; Yu Huang, Ph.D.

M.E. | Computer Engineering — Sept 2018 – Dec 2019
School of Engineering, Vanderbilt University — Nashville, TN

- Coursework in computer architecture, machine learning, and software systems.

B.S. | Computer Science — Aug 2014 – Jun 2018
School of Computer Science, Donghua University — Shanghai, China

ACADEMIC AND RESEARCH APPOINTMENTS

Postdoctoral Research Fellow — Jul 2025 – Present
Department of Biomedical Informatics, Vanderbilt University Medical Center — Nashville, TN

- Develop longitudinal clinical informatics methods that integrate EHR narratives and structured clinical data to measure evolving care states and predict clinically meaningful transitions.
- Conduct interdisciplinary research in clinical NLP, trustworthy AI, longitudinal prediction, and dementia caregiving, with emphasis on evidence provenance, calibration, uncertainty, and prospective evaluation.

Graduate Research Assistant — Jun 2019 – Jun 2025
Vanderbilt University (The SPHERE Lab) — Nashville, TN

- Conducted research at the intersection of biomedical informatics, clinical NLP, machine learning, and social computing using EHRs, patient-reported data, and online caregiver communities.
- Led first-author projects on cancer medication discontinuation and Alzheimer's caregiving communities, combining observational health data with machine learning and computational social science methods.

- Studied public responses to human gene editing across Sina Weibo, Twitter, Reddit, and YouTube using structural topic modeling and annotated stance comparisons.
- Examined how platform and cultural context shape ethical concerns, expectations of disease prevention, and support or opposition to genetic technologies.

Doctoral Research Intern May 2024 – Aug 2024
American Institutes for Research (AIR), Institute of Education Sciences (IES) Arlington, VA

- Designed and evaluated machine learning and NLP models to characterize student response-change behavior in large-scale educational assessment data.
- Collaborated with psychometricians and data scientists to connect model outputs with scoring, feedback, and assessment analytics workflows.

RESEARCH EXPERIENCE

Dementia Care and Caregiving

Longitudinal Care-State Informatics Jan 2026 – Present

- Develop an auditable longitudinal care-state representation from fragmented EHR narratives, organizing changes in patient needs, caregiving capacity, and care arrangements while preserving source, provenance, and temporal context.
- In preliminary analyses, reconstructed 343,884 care episodes from 17,703 patients and evaluated five prototype care-state domains; longitudinal-history models improved precision-recall performance over current-state-only models across all five prototype transitions, with a median relative AUPRC gain of 36%.

Caregiver–AI Communication and RAG System Design Sep 2024 – Dec 2025

- Designed a modular retrieval-augmented generation (RAG) pipeline with domain-grounded retrieval, role-aware generation, and response evaluation components for dementia caregiving support.
- Developed a comparative evaluation framework for clinician, peer caregiver, ChatGPT, and CareGPT responses across emotional, informational, safety, and communication-support dimensions.
- Implemented CareGPT with LangChain and vector stores; benchmarked caregiver, clinician, ChatGPT, and CareGPT responses across 28 emotional and informational support metrics.

Community Engagement Dynamics in ADRD Forums Apr 2024 – Apr 2025

- Built a causal-comparative analysis pipeline integrating propensity score matching, topic modeling, sentiment features, and linguistic markers to characterize drivers of sustained engagement in caregiver forums.
- Analyzed large-scale discussions from TalkingPoint and ALZConnected to quantify how topic-initiator content and emotional framing shape downstream community participation.
- Analyzed over 20,000 posts from TalkingPoint and ALZConnected using LDA topic modeling and psycholinguistic features to contrast threads with and without sustained engagement.

Emotional Trajectories and Support Patterns in ADRD Communities May 2023 – Feb 2024

- Developed a longitudinal emotion-analysis pipeline combining lexicon-based affect scoring, topic modeling, and temporal trajectory analysis to track emotional support across caregiver discussions.
- Identified recurrent emotional progressions and communication archetypes, including informational requests, reassurance exchanges, crisis narratives, and peer validation.
- Characterized specific affect transitions, including guilt to appreciation, within longitudinal caregiver narratives.

Caregiving Role Differences in Online Interaction Sep 2022 – Jul 2023

- Designed comparative behavioral analyses of spousal and adult-child caregivers using sentiment analysis, text-length normalization, and temporal engagement metrics.
- Quantified differences in emotional expression, information seeking, and coping discussions across relationship-based caregiver subgroups.
- Compared reply frequency, linguistic patterns, and engagement depth, characterizing emotional expression and peer empathy among spousal caregivers and factual information seeking among adult-child caregivers.

Lived Experiences of Dementia Caregivers on Reddit

Jun 2021 – May 2022

- Constructed a scalable data collection and preprocessing pipeline for dementia-related Reddit posts using subreddit filtering, data cleaning, and manual content curation.
- Applied topic modeling and structured qualitative coding to characterize caregiver burden, family conflict, emotional exhaustion, and end-of-life decision-making.
- Collected and screened 12,000 dementia-related Reddit caregiving posts, combining subreddit filters, LDA topic modeling, and qualitative theme coding.

Mental Health and V-CARES

V-CARES 1.0: Mental Health Data Generation, Annotation, and Hallucination Detection

Jan 2025 – Jun 2026

- Developed synthetic mental health prompts using a multi-stage LLM pipeline with structured prompt elements, controlled regeneration, and screening for safety, duplication, and coverage.
- Constructed prompt–response pairs and annotation workflows, incorporating concern taxonomies, stakeholder labels, response quality, safety, and confidence-based review.
- Developed hallucination-detection models and evaluated prompt-element-level risks for hallucinations and omissions in mental health LLM responses.

Phase 1 completed; V-CARES Phase 2 is underway.

Breast Cancer

Genomic and Biospecimen-Linked Mammography Cohorts

2024 – Present

- Designed data integration pipelines linking population-scale mammography encounters with whole genome sequencing data and BioVU plasma indicators for multimodal breast cancer research.
- Identified analytic subcohorts with paired WGS–mammography data and available biospecimens, harmonizing metadata across imaging phenotypes, genomic datasets, and plasma resources.
- Established reusable infrastructure for phenotype–genotype association studies, biomarker discovery, and multimodal breast cancer risk modeling.
- Linked 1.6 million mammography encounters with whole-genome sequencing records and BioVU plasma indicators across more than 174,000 screened individuals; identified over 2,000 patients with paired WGS–mammography data and over 6,000 with plasma biospecimens.
- Genomic linkage work identified 2,125 patients with both WGS and mammography records; a plasma-linked cohort retained 6,294 of 174,376 screened mammography patients.
- Built a mammography-based clinical cohort of 157,983 women, integrating cancer and family history, menopausal status, hormone-replacement exposure, age at first birth, and alcohol and smoking data into harmonized subcohorts.

Mammography Note Extraction and Hormone-Therapy Flagging

Apr 2024 – Present

- Developed hybrid information extraction pipelines combining regular expressions and GPT-based LLMs to assign BI-RADS and breast density labels from large-scale mammography notes.
- Benchmarked LLM-assisted extraction against rule-based baselines to improve handling of nuanced, context-dependent, and inconsistently documented imaging descriptions.
- Created hormone-modulating therapy exposure indicators from structured EHR drug records to support imaging–medication interaction analyses and risk modeling.
- Applied information extraction to 1.1 million mammography notes and created hormone-modulating medication exposure flags for more than 176,000 patients.

Cancer Medication Discontinuation

LLM-Based Prediction and Explainability

Jan 2025 – Present

- Extended medication discontinuation modeling to evaluate GPT-4o in zero-shot, few-shot, and fine-tuned settings using text-encoded EHR summaries and structured clinical context.
- Developed evaluation workflows for predictive performance, explanation quality, reasoning fidelity, and error patterns in LLM-based clinical risk prediction tasks.

Multimodal Cohort Modeling for Oral Anti-Cancer Medications

Jun 2023 – Present

- Designed a specialty pharmacy-linked cohort for oral anti-cancer medications by integrating prescription refill patterns, structured EHR data, and patient-reported discontinuation questionnaires.

- Developed comparative machine learning pipelines using logistic regression, random forest, support vector machines, k-nearest neighbors, gradient boosting, and SHAP-based attribution.
- Linked 4,184 medication records from 2,732 patients to refill, EHR, and patient-reported information; examined demographic, clinical, and socioeconomic features with SHAP attribution.

Hormonal Therapy Discontinuation Prediction from Structured EHRs Jun 2020 – Dec 2022

- Constructed a breast cancer cohort using structured EHR features, including conditions, medications, laboratory tests, and procedures, to model hormonal therapy discontinuation.
- Implemented and evaluated multiple machine learning models, showing that streamlined feature sets can perform comparably to full-feature models under realistic missingness patterns.
- Constructed a cohort of 2,251 hormone-receptor-positive breast cancer patients and examined advanced stage, secondary malignancies, heart failure, relapse-related procedures, and biomarkers as discontinuation predictors.

Clinical NLP and EHR Representation

Auditable Concept Construction from Clinical Free Text 2026

- Studied an agentic workflow that derives task-specific concept taxonomies and weak labels from a bounded text sample, routes local extractors over clinical corpora, and assembles an auditable representation for prediction.
- Evaluated count- and recency-based longitudinal concept features for medication discontinuation and 30-day readmission, retaining source evidence for interpretation and audit.

Biological Aging

Estimating Phenotypic Aging from EHR Biomarkers using Modified PheAge Jan 2024 – Jun 2024

- Adapted the PhenoAge biological aging estimator for use in EHRs by refitting the model with 8 available biomarkers due to high CRP missingness.
- Used NHANES-based validation to confirm near-perfect correlation (r = 0.995) between original and modified formulas.
- Extracted and cleaned measurements from >400,000 patients, joined with condition data to characterize aging-related disease burdens.
- Study supports applications in aging prediction, clinical deterioration, and intervention timing.

Education and Assessment

Student Response Changes in NAEP Mathematics Assessment May – Aug 2024

- Designed and evaluated NLP and machine learning approaches to characterize changes in constructed responses during the American Institutes for Research internship.
- Connected changes in grammar, structure, and meaning with automated scoring and student performance.

PARTICIPATION IN FUNDED RESEARCH

ARPA-H V-CARES
Mental-health LLM evaluation and safety
Role: research contributor.
NIH / National Cancer Institute – R37CA237452
Anti-cancer medication discontinuation research in Dr. Zhijun Yin's project
Role: research contributor.

PUBLICATIONS

Journal Articles

1. **Congning Ni**, Bhashithe Abeysinghe, and Juanita Hicks. (2025). *Decoding Student Insights: Analyzing Response Change in NAEP Mathematics Constructed Response Items. International Electronic Journal of Elementary Education*, 17(2), 237–251.
2. Qingyuan Song, **Congning Ni**, Jeremy L. Warner, Qingxia Chen, Lijun Song, S. Trent Rosenbloom, Bradley A. Malin, and Zhijun Yin. (2024). *Optimizing word embeddings for small datasets: a case study on patient portal messages from breast cancer patients. Scientific Reports*, 14, 16117. doi:10.1038/s41598-024-66319-z.

3. **Congning Ni**, Qingyuan Song, Qingxia Chen, Lijun Song, Patricia Commiskey, Lauren Stratton, Bradley Malin, and Zhijun Yin. (2024). *Sentiment Dynamics Among Informal Caregivers in Web-Based Alzheimer Communities: Systematic Analysis of Emotional Support and Interaction Patterns*. *JMIR Aging*, 7, e60050.
4. Leyao Wang, Zhiyu Wan, **Congning Ni**, Qingyuan Song, Yang Li, Ellen Wright Clayton, Bradley A. Malin, and Zhijun Yin. (2024). *Applications and Concerns of ChatGPT and Other Conversational Large Language Models in Health Care: Systematic Review*. *Journal of Medical Internet Research*, 26, e22769.
5. Zhijun Yin, Lauren Stratton, Qingyuan Song, **Congning Ni**, Lijun Song, Patricia Commiskey, Qingxia Chen, Monica Moreno, Sam Fazio, and Bradley A. Malin. (2024). *Perceptions and Utilization of Online Peer Support Among Informal Dementia Caregivers: Survey Study*. *JMIR Aging*, 7(1), e55169.
6. **Congning Ni**, Qingyuan Song, Bradley Malin, Lijun Song, Patricia Commiskey, Lauren Stratton, and Zhijun Yin. (2023). *Examining Online Behaviors of Adult-Child and Spousal Caregivers for People Living with Alzheimer Disease or Related Dementias: Comparative Study in an Open Online Community*. *Journal of Medical Internet Research*, 25, e48193.
7. Yongtai Liu, Zhijun Yin, **Congning Ni**, Chao Yan, Zhiyu Wan, and Bradley A. Malin. (2023). *Examining Rural and Urban Sentiment Difference in COVID-19-Related Topics on Twitter: Word Embedding-Based Retrospective Study*. *Journal of Medical Internet Research*, 25, e42985.
8. **Congning Ni**, Zhiyu Wan, Chao Yan, Yongtai Liu, Ellen Wright Clayton, Bradley Malin, and Zhijun Yin. (2022). *The Public Perception of the #GeneEditedBabies Event Across Multiple Social Media Platforms: Observational Study*. *Journal of Medical Internet Research*, 24(3), e31687.
9. Yongtai Liu, Zhijun Yin, Zhiyu Wan, Chao Yan, Weiyi Xia, **Congning Ni**, Ellen Wright Clayton, Yevgeniy Vorobeychik, Murat Kantarcioglu, and Bradley A. Malin. (2022). *Implicit Incentives Among Reddit Users to Prioritize Attention Over Privacy and Reveal Their Faces When Discussing Direct-to-Consumer Genetic Test Results: Topic and Attention Analysis*. *JMIR Infodemiology*, 2(2), e35702.

Conference and Workshop Papers – Published or Accepted

1. Qingyuan Song, **Congning Ni**, Weixin Liu, Yike Li, Bradley A. Malin, and Zhijun Yin. (2026). *CPT/HCPCS Code Recommendation from Clinical Notes: A Comparative Evaluation of AI Methods*. IEEE BigData 2026, 9th Special Session on Healthcare Data. Accepted.
2. **Congning Ni**, Weixin Liu, Qingyuan Song, Matthew Murrow, Bradley A. Malin, and Zhijun Yin. (2026). *From Clinical Free Text to Auditable Concepts: An Agentic Framework for Interpretable Prediction*. WISE 2026, 3rd International Workshop on AI-for-Good. Accepted.
3. Mingyang Jiang, **Congning Ni**, Weixin Liu, and Zhijun Yin. (2026). *Characterizing Treatment-Context Medication Evidence Across Clinic Notes and Structured EHR Medication History*. IEEE BIBM 2026 Workshop. Accepted; revision requested.
4. Weixin Liu, **Congning Ni**, Qingyuan Song, Susannah L. Rose, Murat Kantarcioglu, Bradley A. Malin, and Zhijun Yin. (2026). *Coverage-Controlled Preference Mining from Noisy Claim Verification for Evidence-Grounded Generation*. Asian Conference on Machine Learning (ACML 2026). Accepted.
5. Weixin Liu, **Congning Ni**, Shelagh A. Mulvaney, Susannah L. Rose, Murat Kantarcioglu, Bradley A. Malin, and Zhijun Yin. (2026). *MHGraphBench: Knowledge Graph-Grounded Benchmarking of Mental Health Knowledge in Large Language Models*. Proceedings of the Fifth Workshop on Generation, Evaluation and Metrics (GEM), 393–409. doi:10.18653/v1/2026.gem-main.38.
6. **Congning Ni**, Jinkyung Katie Park, Yang Li, Sarvech Qadir, Sean S. Huang, James S. Powers, Julia A. Hiner, Vania Leung, Patricia Commiskey, Lijun Song, Qingxia Chen, Pamela J. Wisniewski, Bradley A. Malin, and Zhijun Yin. (2026). *One Question, Four Voices: How Advice for Alzheimer's Caregiving Differs Between Caregivers, Clinicians, and Large Language Models*. *Proceedings of the ACM on Human-Computer Interaction*, 10(6), Article CSCW132. doi:10.1145/3816980.
7. Sarvech Qadir, **Congning Ni**, Mihir Sachin Vaidya, Hyeyoung Ryu, Shelagh Mulvaney, Murat Kantarcioglu, Laurie Lovett Novak, Bradley A. Malin, Susannah Leigh Rose, and Zhijun Yin. (2026). *Beyond the Response: Examining Reasoning and Execution Fidelity in Large Language Models for Mental Health*. ACM Conference on Fairness, Accountability, and Transparency (FAccT 2026). 6702–6722. doi:10.1145/3805689.3806456.
8. **Congning Ni**, Qingyuan Song, Jeremy L. Warner, Qingxia Chen, Lijun Song, S. Trent Rosenbloom, Autumn Zuckerman, Bridget Lynch, Bradley A. Malin, and Zhijun Yin. (2026). *Making LLM Predictions Interpretable: Fine-Tuning GPT-4o for Early Discontinuation of Cancer Medication*. *Studies in Health Technology and Informatics*, 336, 750–754.
9. **Congning Ni**, Sarvech Qadir, Bryan Steitz, Mihir Sachin Vaidya, Qingyuan Song, Lantian Xia, Shelagh Mulvaney, Siru Liu, Hyeyoung Ryu, Leah Hecht, Amy Bucher, Christopher Symons, Laurie Novak, Susannah L. Rose, Murat Kantarcioglu, Bradley A. Malin, and Zhijun Yin. (2026). *Disentangling Prompt Element Level Risk Factors for Hallucinations and Omissions in Mental Health LLM Responses*. AMIA 2026 Annual Symposium. Accepted full paper.

10. **Congning Ni**, Qingyuan Song, Qingxia Chen, Lijun Song, Patricia Commiskey, Lauren Stratton, Bradley Malin, and Zhijun Yin. (2025). *Catalysts of Conversation: Examining Interaction Dynamics Between Topic Initiators and Commentors in Alzheimer's Disease Online Communities*. In *Proceedings of the ACM Web Conference 2025* (pp. 4913–4924).
11. Qingyuan Song, Jessie Yang, Ndidiamaka Obi, **Congning Ni**, Jeremy L. Warner, Qingxia Chen, Lijun Song, S. Trent Rosenbloom, Bradley A. Malin, and Zhijun Yin. (2025). *Using Large Language Model for Efficient Extraction of Treatment Discontinuation Information: A Study of Online Breast Cancer Community Posts*. In *International Conference on Artificial Intelligence in Medicine* (pp. 379–383). Springer Nature Switzerland.
12. **Congning Ni**, Qingyuan Song, Jeremy L. Warner, Qingxia Chen, Lijun Song, S. Trent Rosenbloom, Bradley A. Malin, and Zhijun Yin. (2024). *Examining Oral Anti-Cancer Medication Continuation Using Questionnaires, Prescription Refills, and Structured Electronic Health Records*. *AMIA Annual Symposium Proceedings*, 865–874.
13. Yi Xin, **Congning Ni**, Qingyuan Song, and Zhijun Yin. (2024). *Fatigue, Pain, and Medication: Mining Online Posts Regarding Rheumatoid Arthritis From Reddit*. *AMIA Annual Symposium Proceedings*, 754–763.
14. Qingyuan Song, Yunfei Hu, **Congning Ni**, and Zhijun Yin. (2023). *The Hidden Patient Connections: Predicting Hormonal Therapy Medication Discontinuation Using Hypergraph Neural Network on Clinical Communications*. *AMIA Summits on Translational Science Proceedings*, 505–514.
15. **Congning Ni**, Bradley Malin, Lijun Song, Angela Jefferson, Patricia Commiskey, and Zhijun Yin. (2022). *"Rough Day... Need a Hug": Learning Challenges and Experiences of the Alzheimer's Disease and Related Dementia Caregivers on Reddit*. *Proceedings of the International AAAI Conference on Web and Social Media*, 16(1), 711–722.
16. **Congning Ni**, Jeremy L. Warner, Bradley A. Malin, and Zhijun Yin. (2022). *Predicting Hormonal Therapy Medication Discontinuation for Breast Cancer Patients Using Structured Data in Electronic Medical Records*. *AMIA Summits on Translational Science Proceedings*, 359–368.

Conference Abstracts and Short Contributions

1. **Congning Ni**, Eric Chen, Lauren Stratton, Patricia Commiskey, Lijun Song, Qingxia Chen, Pamela Wisniewski, Bradley Malin, and Zhijun Yin. (2025). *Examining Online Caregiving Discussions Across Racial Groups of Informal Alzheimer's Disease Caregivers*. *Studies in Health Technology and Informatics*, 329, 2094–2095.
2. **Congning Ni**, Jonathan Hughes, Andrew Nam, and Zhijun Yin. (2021). *Identifying Posts from Reddit Talking About ADRD Caring Experiences Using Machine Learning*. *AMIA 2021 Annual Symposium Proceedings*, 1766.

Book Chapter

1. Zhijun Yin, **Congning Ni**, Daniel Fabbri, S. Trent Rosenbloom, and Bradley A. Malin. (2022). *Detecting Personal Health Mentions from Social Media Using Supervised Machine Learning*. In *Personal Health Informatics: Patient Participation in Precision Health* (pp. 247–266). Springer.

Editorial and Other Scholarly Contributions

1. Steffen Staab, Luca Maria Aiello, Yelena Mejova, Oshani Seneviratne, Cornelia Sindermann, Thilo Hagendorff, Simone Brandes, Juliane Rohlmann, Marc Spaniol, Katharina Kinder-Kurlanda, **Congning Ni**, and others. (2024). *Web Science 2024: Conference Chairs' Welcome Message*. Proceedings of the 16th ACM Web Science Conference (WebSci 2024), III–X.

PREPRINTS AND SUBMITTED MANUSCRIPTS

1. **Congning Ni**, Lantian Xia, Lijun Song, Qingxia Chen, Sean S. Huang, Bradley Malin, and Zhijun Yin. (2026). *Online Peer Support Use Among Informal Caregivers of People Living With Alzheimer Disease or Related Dementias: National Cross-Sectional Web-Based Survey*. medRxiv. Under review at JMIR Aging.
2. Weixin Liu, Juming Xiong, **Congning Ni**, Yanfan Zhu, Xingtao Lin, Bradley A. Malin, and Zhijun Yin. (2026). *DRIFT: Direct-Recursive Intervention-Conditioned Forecasting of ICU Physiological Trajectories*. arXiv.
3. Juming Xiong, Weixin Liu, Kevin H. Guo, **Congning Ni**, Junchao Zhu, Chongyu Qu, Chao Yan, Katherine Brown, Avinash Baidya, Xiang Gao, Bradley A. Malin, and Zhijun Yin. (2026). *CoRA: Confidence–Rationale Alignment for Reliable Chain-of-Thought Reasoning*. arXiv.
4. Weixin Liu, Bowen Qu, Juming Xiong, **Congning Ni**, Bradley A. Malin, and Zhijun Yin. (2026). *Vectors Are Not Neutral: Sensitive-Information Inference from Exported LLM Representations in Summarization*. *arXiv*.
5. Juming Xiong, Kevin Guo, **Congning Ni**, Weixin Liu, Chao Yan, Katherine Brown, Avinash Baidya, Xiang Gao, Bradley Malin, and Zhijun Yin. (2026). *Learning When to Sample: Confidence-Aware Selective Sampling for Efficient Chain-of-Thought Reasoning*. arXiv.

RESEARCH SOFTWARE AND REPRODUCIBILITY

Agentic Clinical NLP Pipeline
Public research-code repository
Code for literature-grounded concept construction, budgeted weak labeling, routed clinical extraction, and auditable concept-based prediction.

Cancer Medication Discontinuation Reproducibility Package
Public code and prompt templates
Code-only package for traditional ML baselines, GPT-4o prompting and fine-tuning inputs, and attribution aggregation.

TEACHING AND ACADEMIC SERVICE

Reviewer and Program Committee Member Mar 2020 – Present
Journals and Conferences in Computer Science, HCI, and Biomedical Informatics

- Serve as a reviewer and program committee member for venues in computer science, human-computer interaction, biomedical informatics, and digital health.
- Reviewed over 30 manuscripts and conference papers for venues including WebSci, ICTAI, AMIA, JMIR, ICWSM, CHI, CSCW, and related journals and conferences.
- Evaluate research on machine learning for health, clinical NLP, social computing, AI ethics, and trustworthy AI in clinical and caregiving contexts.

Guest Lecturer Spring 2021 – Spring 2024
Department of Computer Science, Vanderbilt University Nashville, TN

- Delivered graduate-level guest lectures on natural language processing, transformer architectures, and vector-based semantic representations for health-related text.
- Presented topic modeling and representation methods, including structured topic models, BERTopic, LDA, NMF, BERT, GPT, and semantic embeddings, with health-related text examples.
- Introduced inverse probability of treatment weighting and propensity score matching for observational data analysis.
- Integrated research-driven case studies, datasets, and analytical workflows into AI-for-health and NLP course materials.

RESEARCH MENTORING

Vanderbilt University and Vanderbilt University Medical Center

- Mentored high-school, undergraduate, master's, and early-stage PhD trainees on clinical NLP, EHR and social-media research, online health communities, and AI for caregiving.
- Guided research-question formulation, literature review, study and analysis-pipeline design, interpretation of findings, and manuscript development.
- Mentoring collaborations contributed to coauthored peer-reviewed papers, posters, and abstracts.

HONORS AND RESEARCH RECOGNITION

Gary Marsden Travel Award Sep 2026
ACM SIGCHI

CSCW 2026 Impact Recognition Sep 2026
For "One Question, Four Voices: How Advice for Alzheimer's Caregiving Differs Between Caregivers, Clinicians, and Large Language Models"

Vanderbilt Health News Research Highlight Jul 2026
Coverage of research comparing GPT-4o and traditional ML for cancer medication discontinuation.

Graduate School Travel Award 2022 – 2025
Vanderbilt University Graduate School; received annually

INVITED TALKS AND SEMINARS

Vanderbilt Interdisciplinary Alzheimer's Disease Journal Club 2026 Vanderbilt University Medical Center
Invited presenter and discussion leader on early Alzheimer's disease prediction using longitudinal EHR data; hosted by VMAC and VADRC